Model Predictive Control

Chapter 8: Robust MPC for Linear Systems
Part I

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Outline

1. Uncertainty Models
2. Tools: Robust Invariance, Set Difference & Sum
3. Impact of Bounded Additive Noise
4. Robust Open-Loop MPC

Robust MPC: Motivation

1. MPC relies on a model, but models are far from perfect

2. Noise and model inaccuracies can cause:
   - Constraint violation
   - Sub-optimal behaviour

3. Persistent noise prevents the system from converging to a single point

4. Can incorporate some noise models into the MPC formulation
   - Solving the resulting optimal control problem can be extremely difficult

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MPC vs The Real World

Predictive control assumption:

$$x(k+1) = g(x(k), u(k))$$

- System evolves in a predictable fashion

The real world:

$$x(k+1) = g(x(k), u(k), w(k); \theta)$$

- Random noise w changes the evolution of the system
- Model structure is unknown
- Unknown parameters θ impact the dynamics

(w changes with time, θ is unknown, but constant)

This lecture: What can we hope to do in this (real) situation?

Goals of Robust Constrained Control

Uncertain Constrained System

$$x(k+1) = g(x(k), u(k), w(k); \theta) \qquad x, u \in \mathcal{X}, \mathcal{U} \quad w \in \mathcal{W} \quad \theta \in \Theta$$

Design control law $u(k) = \kappa(x(k))$ such that the system:

1. Satifies constraints : $\{x(k)\} \subset \mathcal{X}$, $\{u(k)\} \subset \mathcal{U}$ for all disturbance realizations
2. Is stable: Converges to a neighbourhood of the origin
3. Optimizes (expected/worst-case) "performance"
4. Maximizes the set $\{x(0) \,|\, \text{Conditions 1-3 are met}\}$

Meeting these goals requires some knowledge/assumptions about the random values w and θ.

Examples of Common Uncertainty Models

- **Measurement / Input Bias:** (last lecture)

$$g(x(k), u(k), w(k); \theta) = \tilde{g}(x(k), u(k)) + \theta$$

  θ unknown, but constant

- **Linear Parameter Varying System:**

$$g(x(k), u(k), \theta(k)) = \left(\sum_{j=0}^{n_\theta} \theta_j(k) A_j \right) x(k) + \left(\sum_{j=0}^{n_\theta} \theta_j(k) B_j \right) u(k)$$

  time-varying parameters $\theta(k)$ describe a convex combination of A_j, B_j with $\mathbf{1}^T \theta(k) = 1$, $\theta(k) \geq 0$

- **Additive Stochastic Noise**

$$g(x(k), u(k), w(k); \theta) = Ax(k) + Bu(k) + w(k)$$

  Distribution of w known

Examples of Common Uncertainty Models

Additive Bounded Noise

$$g(x(k), u(k), w(k); \theta) = Ax(k) + Bu(k) + w(k) \ , \quad w \in \mathcal{W}$$

A, B known, w unknown and changing at each sampling instance

- Dynamics are linear, but impacted by random, bounded noise at each time step
- Can model many nonlinearities in this fashion, but often a conservative model
- The noise is *persistent*, i.e., it does not converge to zero in the limit

We will focus on uncertainty models of this form in this course.

Learning Objectives

- Understand concept of robust invariance, learn algorithm to compute robust invariant sets
- Understand impact of additive noise on system trajectory
- Reformulate constraints to ensure robust constraint satisfaction
- Formulate robust open-loop MPC problem

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Reminder: Invariance

Constraint satisfaction, for an **autonomous** system $x(k+1) = g(x(k))$, or **closed-loop** system $x(k+1) = g(x(k), \kappa(x(k)))$ for a **given** controller κ.

Positive Invariant set
A set $\mathcal{O}$ is said to be a positive invariant set for the autonomous system $x(k+1) = g(x(k))$ if $$x(k) \in \mathcal{O} \;\; \Rightarrow \;\; x(k+1) \in \mathcal{O} \;\; , \;\; \forall k \in \{0, 1, \dots\}$$

If we have an invariant set $\mathcal{X}_f \subseteq \mathcal{X}$ and $\kappa(\mathcal{X}_f) \subseteq \mathcal{U}$, then it provides a set of initial states from which the trajectory will never violate the system constraints if we apply the controller κ.

Robust Invariance

Robust constraint satisfaction, for **autonomous** system
$x(k+1) = g(x(k), w(k))$,
or **closed-loop** system $x(k+1) = g(x(k), \kappa(x(k)), w(k))$ for a **given** controller κ.

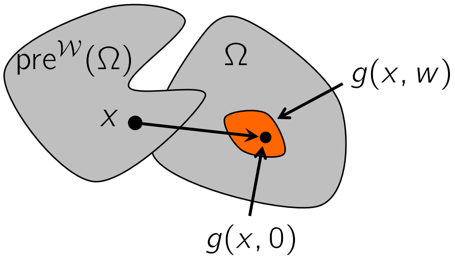
Robust Positive Invariant set
A set $\mathcal{O}^{\mathcal{W}}$ is said to be a robust positive invariant set for the autonomous system $x(k+1) = g(x(k), w(k))$ if $$x \in \mathcal{O}^{\mathcal{W}} \quad \Rightarrow \quad g(x, w) \in \mathcal{O}^{\mathcal{W}} \quad , \quad \text{for all } w \in \mathcal{W}$$

Robust Pre-Sets

Robust Pre Set

Given a set Ω and the dynamic system $x(k+1) = g(x(k), w(k))$, the **pre-set** of Ω is the set of states that evolve into the target set Ω in one time step *for all values of the disturbance* $w \in \mathcal{W}$:

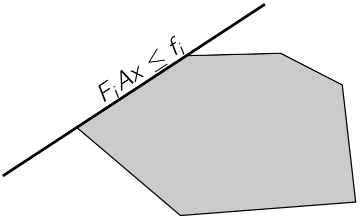
$$\text{pre}^{\mathcal{W}}(\Omega) := \{x \,|\, g(x,w) \in \Omega \text{ for all } w \in \mathcal{W}\}$$



Computing Robust Pre-Sets for Linear Systems

Goal: Given the system $g(x(k), w(k)) = Ax(k) + w(k)$, and the set $\Omega := \{x \,|\, Fx \leq f\}$, compute $\mathrm{pre}^{\mathcal{W}}(\Omega)$.

$$\mathrm{pre}^{\mathcal{W}}(\Omega) = \{x \,|\, Ax + w \in \Omega,\ \forall w \in \mathcal{W}\} = \{x \,|\, FAx + Fw \leq f,\ \forall w \in \mathcal{W}\}$$



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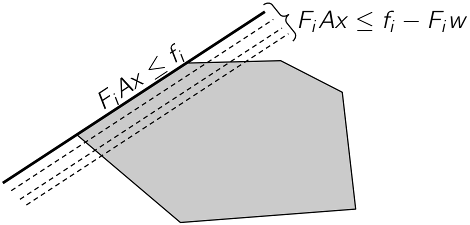

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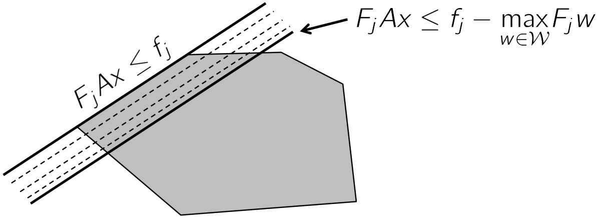

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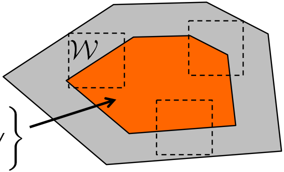
$$\left\{ x \,\middle|\, FAx \le f - \max_{w \in \mathcal{W}} Fw \right\}$$



$$\mathrm{pre}^{\mathcal{W}}(\Omega) = \left\{ x \,\middle|\, FAx \le f - \max_{w \in \mathcal{W}} Fw \right\} = \{x \mid FAx \le f - h_{\mathcal{W}}(F)\}$$

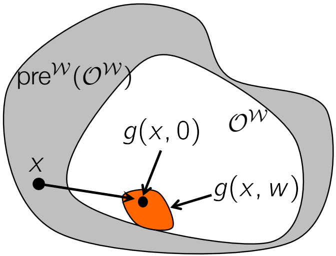
where $h_{\mathcal{W}}$ is the *support function*.

Robust Invariant Set Conditions

Theorem: Geometric condition for robust invariance

A set $\mathcal{O}^{\mathcal{W}}$ is a robust positive invariant set if and only if

$$\mathcal{O}^{\mathcal{W}} \subseteq \mathrm{pre}^{\mathcal{W}}(\mathcal{O}^{\mathcal{W}})$$



Computing Robust Invariant Sets

Conceptual Algorithm to Compute Robust Invariant Set

Input: g, $\mathcal{X}$, $\mathcal{W}$
Output: $\mathcal{O}_\infty^\mathcal{W}$

```
Ω_0 ← X
loop
    Ω_{i+1} ← pre^W(Ω_i) ∩ Ω_i
    if Ω_{i+1} = Ω_i then
        return O_∞^W = Ω_i
    end if
end loop
```

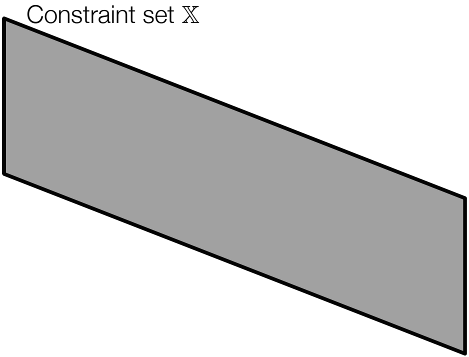
This is the same as for the nominal case, with $\mathrm{pre}(\Omega)$ replaced by $\mathrm{pre}^\mathcal{W}(\Omega)$.

Computing Robust Invariant Sets

$$x(k+1) = (A+BK)x(k) + w(k) \qquad A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \qquad B = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}$$

$$\mathcal{X} = \{x \mid \|x\|_\infty \leq 5,\ \|Kx\|_\infty \leq 1\} \qquad \mathcal{W} = \{w \mid \|w\|_\infty \leq 0.3\}$$

K is the LQR controller for $Q = 0.1I$, $R = 1$



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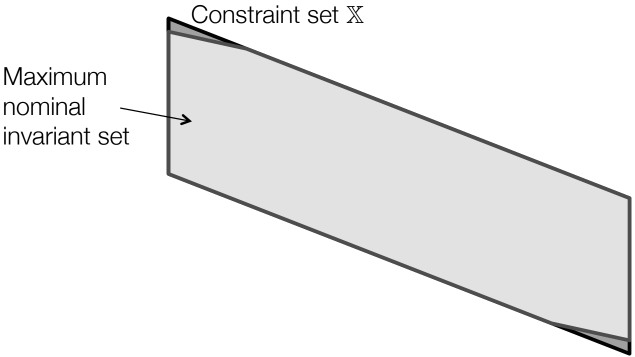

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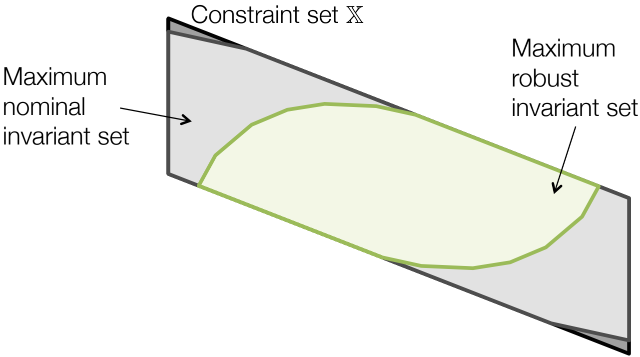

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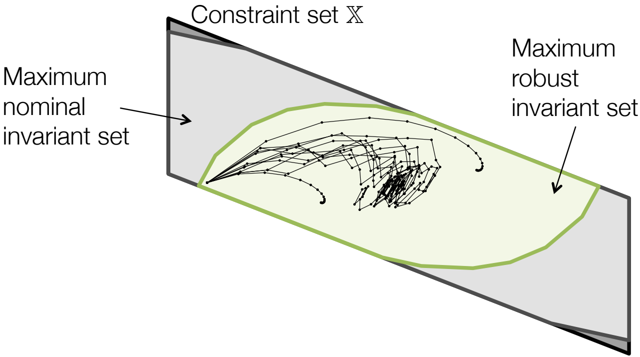

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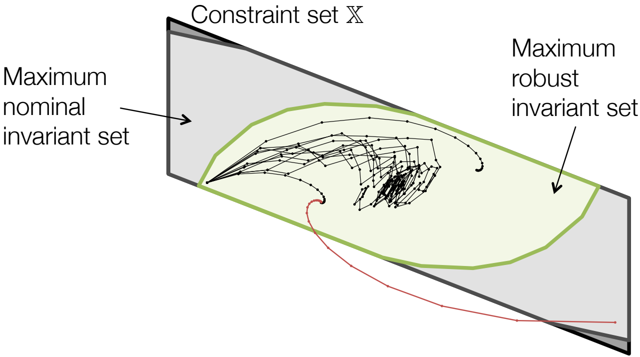


Minkowski Sum and Pontryagin Difference

Minkowski Sum

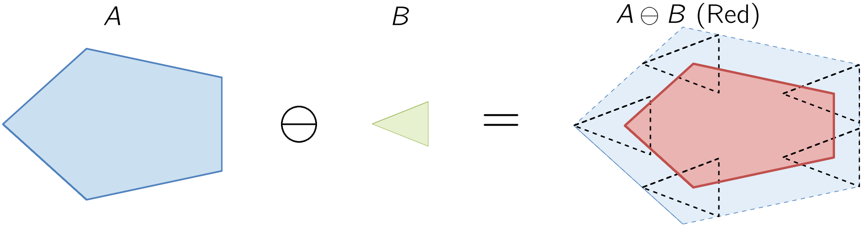
Let A and B be subsets of $\mathbb{R}^n$. The Minkowski Sum is

$$A \oplus B := \{x + y \,|\, x \in A, y \in B\}$$

Pontryagin Difference

Let A and B be subsets of $\mathbb{R}^n$. The Pontryagin Difference is

$$A \ominus B := \{x \,|\, x + e \in A \; \forall e \in B\}$$



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Goals of Robust Constrained Control

Uncertain constrained linear system
$$x(k+1) = Ax(k) + Bu(k) + w(k) \qquad x, u \in \mathcal{X}, \mathcal{U} \qquad w \in \mathcal{W}$$

Design control law $u(k) = \kappa(x(k))$ such that the system:

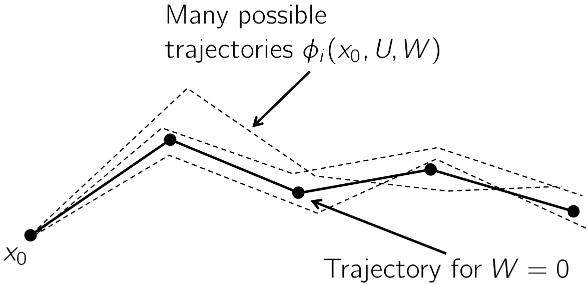
1. Satifies constraints : $\{x(k)\} \subset \mathcal{X}$, $\{u(k)\} \subset \mathcal{U}$ for all disturbance realizations
2. Is stable: Converges to a neighbourhood of the origin
3. Optimizes (expected/worst-case) "performance"
4. Maximizes the set $\{x(0) \,|\, \text{Conditions 1-3 are met}\}$

Challenge: Cannot predict where the state of the system will evolve
We can only compute a set of trajectories that the system *may* follow

Idea: Design a control law that will satisfy constraints and stabilize the system *for all possible disturbances*

Uncertain State Evolution

Given the current state $x(0)$, the model $x(k+1) = Ax(k) + Bu(k) + w(k)$ and the set $\mathcal{W}$, where can the state be i steps in the future?



Define $\phi_i(x_0, U, W)$ as the state that the system will be in at time i if the state at time zero is x_0, we apply the input $U := \{u_0, \ldots, u_{N-1}\}$ and we observe the disturbance $W := \{w_0, \ldots, w_{N-1}\}$.

Uncertain State Evolution

Nominal system

$$x(k+1) = Ax(k) + Bu(k)$$

$$x_1 = Ax_0 + Bu_0$$
$$x_2 = A^2x_0 + ABu_0 + Bu_1$$
$$\vdots$$
$$x_i = A^i x_0 + \sum_{j=0}^{i-1} A^j B u_{i-1-j}$$

Uncertain system

$$x(k+1) = Ax(k) + Bu(k) + w(k) \ , w \in \mathcal{W}$$

$$\phi_1 = Ax_0 + Bu_0 + w_0$$
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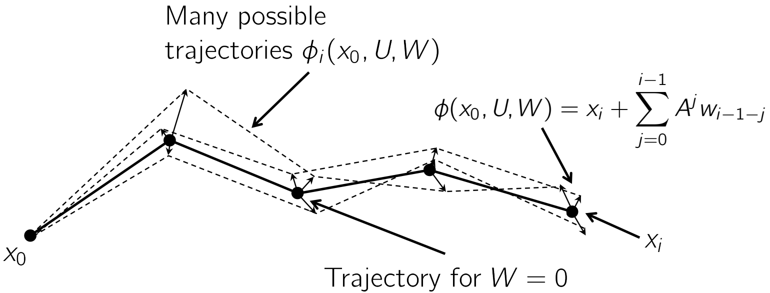
$$\phi_i = A^i x_0 + \sum_{j=0}^{i-1} A^j B u_{i-1-j} + \sum_{j=0}^{i-1} A^j w_{i-1-j}$$
$$\phi_i = x_i + \sum_{j=0}^{i-1} A^j w_{i-1-j}$$

Uncertain evolution is the nominal system + offset caused by the disturbance (Follows from linearity)

Uncertain State Evolution



Goals of Robust Constrained Control

Uncertain constrained linear system
$x(k+1) = Ax(k) + Bu(k) + w(k) \qquad x, u \in \mathcal{X}, \mathcal{U} \qquad w \in \mathcal{W}$

Design control law $u(k) = \kappa(x(k))$ such that the system:

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3. **Optimizes (expected/worst-case) "performance"**
4. Maximizes the set $\{x(0) \,|\, \text{Conditions 1-3 are met}\}$

Defining a Cost to Minimize

Previously, we defined some function that describes a ‘good’ trajectory:

$$J(x_0, U) := \sum_{i=0}^{N-1} l(x_i, u_i) + l_f(x_N)$$

However, there are now many trajectories that *may* occur, depending on the disturbance W.

The cost is now a function of the disturbance seen, and therefore *each possible trajectory has a different cost*:

$$J(x_0, U, W) := \sum_{i=0}^{N-1} l(\phi_i(x_0, U, W), u_i) + l_f(\phi_N(x_0, U, W))$$

Need to ‘eliminate’ the dependence on W.

Defining a Cost to Minimize

Several common options:

- Minimize the expected value (requires some assumption on the distribution)

$$J_N(x_0, U) := E\left[J(x_0, U, W)\right]$$

- Take the worst-case

$$J_N(x_0, U) := \max_{W \in \mathcal{W}^N} J(x_0, U, W)$$

- Take the nominal case

$$J_N(x_0, U) := J(x_0, U, 0)$$

In this lecture we will assume the nominal case for simplicity.

Goals of Robust Constrained Control

Uncertain constrained linear system
$x(k+1) = Ax(k) + Bu(k) + w(k) \qquad x, u \in \mathcal{X}, \mathcal{U} \qquad w \in \mathcal{W}$

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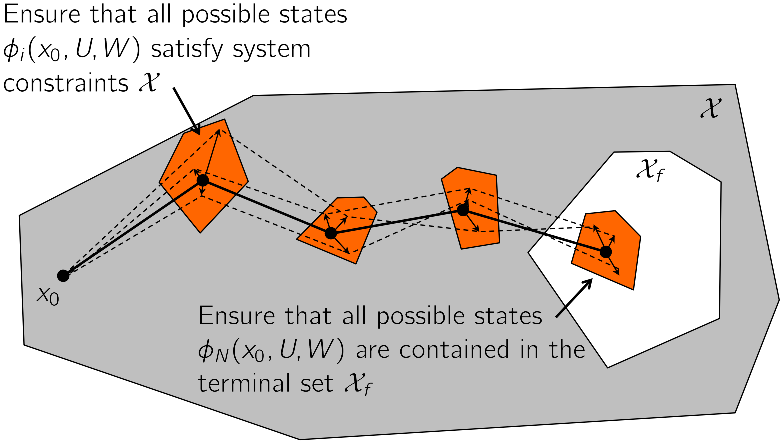
Robust Constraint Satisfaction



The idea: Compute a set of tighter constraints such that if **the nominal system** meets these constraints, then the uncertain system will too.
We then do MPC **on the nominal system.**

Robust Constraint Satisfaction

Recall: We break the MPC prediction into two parts.

$$\left.\begin{aligned} \phi_{i+1} &= A\phi_i + Bu_i + w_i \\ u_i &\in \mathcal{U} \\ \phi_i &\in \mathcal{X}\ \forall W \in \mathcal{W}^N \end{aligned}\right\}$$

- $i = 0, \ldots, N-1$
- Optimize over control actions $\{u_0, \ldots, u_{N-1}\}$
- Enforce constraints explicitly by imposing $\phi_i \in \mathcal{X}$ and $u_i \in \mathcal{U}$ for all sequences W

$$\left.\begin{aligned} \phi_N &\in \mathcal{X}_f \\ \phi_{i+1} &= A\phi_i + w_i \end{aligned}\right\}$$

- $i = N, \ldots$
- Assume no input $u_i = 0$
- Enforce constraints implicitly by constraining ϕ_N to be in a *robust invariant set* $\mathcal{X}_f \subseteq \mathcal{X}$ and $0 \in \mathcal{U}$ for the system $\phi_{i+1} = A\phi_i + w_i$

- Robustly enforcing constraints of a linear system
- Robustly ensuring constraints of the sequence $\phi_1, \ldots, \phi_{N-1}$

Robust Constraint Satisfaction

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- **Robustly ensuring constraints of the sequence** $\phi_1, \ldots, \phi_{N-1}$

Uncertain State Evolution

Nominal system

$$x(k+1) = Ax(k) + Bu(k)$$

$$x_1 = Ax_0 + Bu_0$$
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Uncertain system

$$x(k+1) = Ax(k) + Bu(k) + w(k)\ ,\ w \in \mathcal{W}$$

$$\phi_1 = Ax_0 + Bu_0 + w_0$$
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$$\phi_i = A^ix_0 + \sum_{j=0}^{i-1} A^j Bu_{i-1-j} + \sum_{j=0}^{i-1} A^j w_{i-1-j}$$
$$\phi_i = x_i + \sum_{j=0}^{i-1} A^j w_{i-1-j}$$

Goal: Ensure $\phi_i \in \mathcal{X}$ for all $W \in \mathcal{W}^N$

Robust Constraint Satisfaction

Goal: Ensure that constraints are satisfied for the MPC sequence.

$$\phi_i(x_0, U, W) = \left\{ x_i + \sum_{j=0}^{i-1} A^j w_{i-1-j} \;\middle|\; W \in \mathcal{W}^i \right\} \subseteq \mathcal{X}$$

Assume that $\mathcal{X} = \{x \,|\, Fx \leq f\}$, then this is equivalent to

$$Fx_i + F \sum_{j=0}^{i-1} A^j w_{i-1-j} \leq f \;\; \forall W \in \mathcal{W}^i$$

We've seen this before while computing the robust pre-set:

$$Fx_i \leq f - \max_{W \in \mathcal{W}^i} F \sum_{j=0}^{i-1} A^j w_{i-1-j} = f - h_{\mathcal{W}^i}\left(F \begin{bmatrix} A^0 & A^1 & \dots & A^{i-1} \end{bmatrix}\right)$$

The support function can be pre-computed offline.
All we're doing is tightening the constraints on the nominal system

Terminal State Constraint

We also need to ensure that the N^{th} state $\phi_N(x_0, U, W)$ is contained in the robust invariant set $\mathcal{X}_f$:

$$\phi_N(x_0, U, W) \subseteq \mathcal{X}_f$$

This is handled in exactly the same fashion.

Robust Constraint Satisfaction

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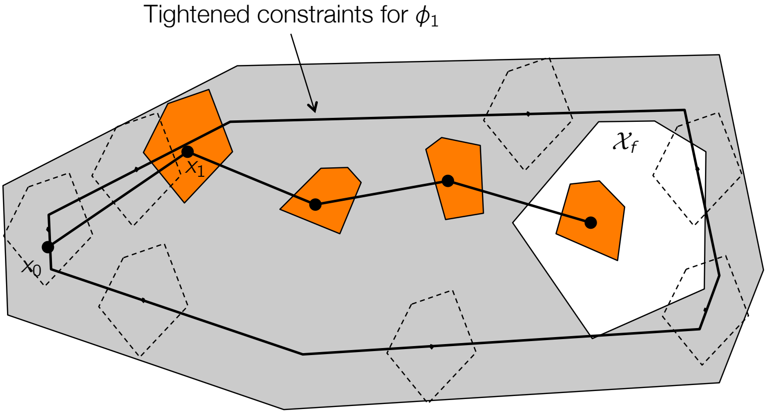

Nominal x_i satisfies tighter constraints $\rightarrow$ Uncertain state does too

Robust Constraint Satisfaction – Notation

Goal: Ensure that constraints are satisfied for the MPC sequence.

$$\phi_i(x_0, U, W) = \left\{ x_i + \sum_{j=0}^{i-1} A^j w_{i-1-j} \;\middle|\; W \in \mathcal{W}^i \right\} \subseteq \mathcal{X}$$

which can be written as

$$\phi_i(x_0, U, W) \in x_i \oplus (\mathcal{W} \oplus A\mathcal{W} \oplus \ldots A^{i-1}\mathcal{W}) \subseteq \mathcal{X}$$

- In order to enforce this condition we require

$$x_i \in \mathcal{X} \ominus (\mathcal{W} \oplus A\mathcal{W} \oplus \ldots A^{i-1}\mathcal{W}),$$

  also denoted as $x_i \in \mathcal{X} \ominus (\bigoplus_{j=0}^{i-1} A^j \mathcal{W})$

- The set $\mathcal{F}_i = \bigoplus_{j=0}^{i-1} A^j \mathcal{W}$ is called **disturbance reachable set**.

  Note: $\mathcal{F}_{i+1} = A\mathcal{F}_i \oplus \mathcal{W}$

Outline

1. Uncertainty Models

2. Tools: Robust Invariance, Set Difference & Sum

3. Impact of Bounded Additive Noise

4. Robust Open-Loop MPC

Putting it Together

Robust Open-Loop MPC

$$
\begin{aligned}
\min_{U} \sum_{i=0}^{N-1} & l(x_i, u_i) + l_f(x_N) \\
\text{subj. to } x_{i+1} &= Ax_i + Bu_i \\
x_0 &= x(k) \\
x_i &\in \mathcal{X} \ominus (\mathcal{W} \oplus A\mathcal{W} \oplus \ldots A^{i-1}\mathcal{W}) \\
u_i &\in \mathcal{U} \\
x_N &\in \mathcal{X}_f \ominus (\mathcal{W} \oplus A\mathcal{W} \oplus \ldots A^{N-1}\mathcal{W})
\end{aligned}
$$

where $\mathcal{X}_f \subseteq \mathcal{X}$ is a robust positive invariant set for the system $x(k+1) = Ax(k) + w(k)$ with $w(k) \in \mathcal{W}$ for all k.

We do **nominal MPC**, but with tighter constraints on the states.

Intuitively: We can be sure that if the nominal system satisfies the tighter constraints, then the uncertain system will satisfy the real constraints.

Goals of Robust Constrained Control

Uncertain constrained linear system
$x(k+1) = Ax(k) + Bu(k) + w(k) \qquad x, u \in \mathcal{X}, \mathcal{U} \qquad w \in \mathcal{W}$

Design control law $u(k) = \kappa(x(k))$ such that the system:

1. Satifies constraints : $\{x(k)\} \subset \mathcal{X}$, $\{u(k)\} \subset \mathcal{U}$ for all disturbance realizations
2. **Is stable: Converges to a neighbourhood of the origin**
3. Optimizes (expected/worst-case) "performance"
4. Maximizes the set $\{x(0) \,|\, \text{Conditions 1-3 are met}\}$

Analysis needs more general stability theory.
($\rightarrow$ Advanced MPC)

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Robust open-loop MPC potentially has a *very* small region of attraction, in particular for unstable systems
Next, we will see techniques introducing feedback.